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PROCUREMENT SPECIFICATION
FF-B-187

INTERNATIONAL INTEREST

SUPERSEDES:

TITLE
BEARING, ROLLER, TAPERED, DOUBLE ROW
OF ROLLERS, NORMAL ANGLE, TWO SINGLE CONES,
ONE DOUBLE CUP, NON-ADJUSTABLE, TYPE 764 (TNA)

MILITARY STANDARD

MS 19088

SHEET 1 OF 6

FED. SUP CLASS
3110

3110-0519

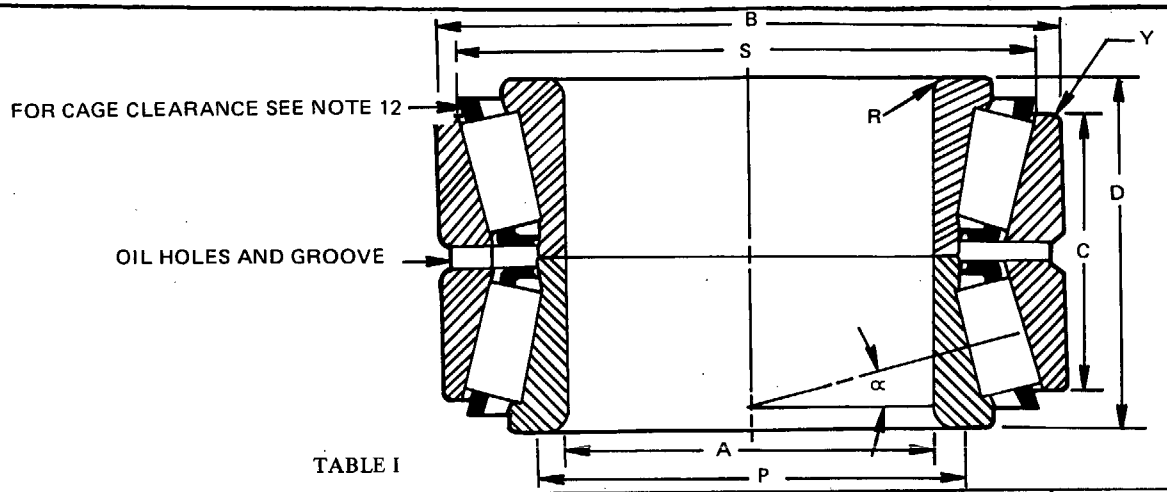


TABLE I

Dash No.	AFBMA (A) bearing number (See Note 11) Cone–Cup	A Bore	B Outside dia.	C Bearing width over cup	D Cones width	R (A) Max. shaft fillet radius	Y (A) Max. housing fillet radius	P S Recommended shoulder (A) diameter: Shaft	Housing	(A) K factor	Basic dynamic load ratings (lb) (A): Radial	Thrust
* 1	NA05075–05185D	0.7500	1.8504	0.9926	1.2500	0.05	0.03	1.00	1.67	1.64	1940	620
* 2	NA08125–08231D	1.2500	2.3125	0.9688	1.2812	0.06	0.02	1.52	2.17	1.23	2440	1040
* 3	NA13687–13621D	1.5000	2.7170	1.5000	1.8124	0.08	0.03	1.83	2.56	1.45	3600	1240
* 4	NA438–432D	1.7500	3.7500	2.0000	2.4376	0.14	0.03	2.24	3.43	2.05	8400	2040
23	NA366–363D	1.9685	3.5433	1.6563	1.9689	0.14	0.03	2.40	3.31	1.83	6300	1810
* 5	NA3780–3729D	2.0000	3.6718	2.0625	2.5625	0.14	0.03	2.52	3.46	1.73	9500	2890
6	NA455–452D	2.000	4.2500	2.1250	2.5626	0.14	0.03	2.56	3.94	1.74	10200	3090
24	NA539–533D	2.1250	4.3750	2.5000	3.1250	0.14	0.06	2.68	3.94	1.97	13100	3500
25	NA385–384D	2.1654	3.9370	1.6875	2.0625	0.14	0.03	2.64	3.66	1.65	6850	2180
* 7	NA558–552D	2.3750	4.8750	2.5000	3.1250	0.14	0.06	2.99	4.53	1.69	15100	4700
26	NA482–472D	2.7500	4.7244	2.1250	2.5626	0.14	0.03	3.27	4.49	1.52	11300	3900

(A) denotes changes. For NOTES see sheets 5 and 6.

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MILITARY STANDARD
MS 19088
SHEET 2 OF 6

Dash No.	AFBMA (A) bearing number (See Note 11) Cone–Cup	A	B	C	D	R	Y	P	S	K factor	Basic dynamic load ratings (lb) (A)	
		Bore	Outside dia.	Bearing width over cup	Cones width	(A) Max. shaft fillet radius	(A) Max. housing fillet radius	Recommended shoulder (A) diameter				
								Shaft	Housing		Radial	Thrust
27	NA484–472D	2.7559	4.7244	2.1250	2.5626	0.14	0.03	3.27	4.49	1.52	11300	3900
28	NA567–563D	2.8750	5.0000	2.5625	3.1875	0.27	0.06	3.70	4.69	1.61	15700	5100
* 8	NA495A–493D	3.0000	5.3750	2.1250	2.7500	0.14	0.03	3.62	5.12	1.31	12500	5000
9	NA659–654D	3.0000	6.0000	3.0000	3.7500	0.14	0.06	3.66	5.55	1.43	19500	7150
*10	NA593–592D	3.5000	6.0000	2.5000	3.2500	0.14	0.03	4.09	5.67	1.32	17900	7100
11	NA759–752D	3.5000	6.3750	3.3750	4.1250	0.14	0.06	4.17	5.91	1.71	26200	8050
29	NA52375–52637D	3.7500	6.3750	2.4375	3.2499	0.14	0.03	4.41	6.06	1.23	18800	8050
30	NA776–773D	3.7500	7.0866	3.3750	4.1250	0.14	0.03	4.49	6.61	1.51	28300	9850
31	NA776–774D	3.7500	7.1250	3.3750	4.1250	0.14	0.06	4.49	6.61	1.51	28300	9850
32	NA691–672D	4.0000	6.6250	2.7500	3.6250	0.14	0.03	4.65	6.30	1.24	21900	9250
33	NA780–773D	4.0000	7.0866	3.3750	4.1250	0.14	0.03	4.69	6.61	1.51	28300	9850
12	NA780–774D	4.0000	7.1250	3.3750	4.1250	0.14	0.06	4.69	6.61	1.51	28300	9850
*13	NA861–854D	4.0000	7.5000	4.0000	5.0000	0.14	0.06	4.72	6.85	1.74	36900	11100
34	NA782–773D	4.1250	7.0866	3.3750	4.1250	0.14	0.03	4.80	6.61	1.51	28300	9850
35	NA782–774D	4.1250	7.1250	3.3750	4.1250	0.14	0.06	4.80	6.61	1.51	28300	9850
14	NA71450–71751D	4.5000	7.5000	3.1875	4.1875	0.14	0.06	5.20	7.13	1.40	30200	11300
15	NA938–932D	4.5000	8.3750	4.6250	5.6250	0.14	0.06	5.28	7.60	1.79	46500	13600
36	NA48291–48220D	5.0000	7.1875	2.8750	3.3750	0.14	0.03	5.55	6.93	1.91	24600	6750
*16	NA798–792D	5.0000	8.1250	3.2500	4.2500	0.14	0.03	5.71	7.80	1.27	32000	13200
17	NA95500–95927D	5.0000	9.2500	4.5000	5.6250	0.14	0.06	5.87	8.54	1.58	52000	17300
*18	NA81550–81963D	5.5000	9.6250	3.1250	4.2500	0.14	0.06	6.34	9.02	1.66	32700	10400
37	NA48686–48620D	5.6250	7.8750	2.8750	3.6876	0.14	0.03	6.22	7.60	1.74	26600	8000
38	NA82576–82951D	5.7500	9.5000	4.1875	5.1875	0.14	0.06	6.54	8.90	1.32	44000	17400
19	NA99600–99102D	6.0000	10.0000	4.3750	5.6250	0.14	0.06	6.85	9.37	1.43	55500	20400

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INTERNATIONAL INTEREST

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ONE DOUBLE CUP, NON-ADJUSTABLE, TYPE 764 (TNA)

MILITARY STANDARD

MS 19088

SHEET 3 OF 6

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Dash No.	AFBMA (A) bearing number (See Note 11) Cone–Cup	A	B	C	D	R	Y	P	S	K factor	Basic dynamic load ratings (lb) (A)	
		Bore	Outside dia.	Bearing width over cup	Cones width	(A) Max. shaft fillet radius	(A) Max. housing fillet radius	Recommended shoulder (A) diameter				
								Shaft	Housing		Radial	Thrust
39	NA67787–67720D	6.8750	9.7500	3.3125	4.0625	0.14	0.03	7.56	9.45	1.33	37100	14700
40	NA87700–87112D	7.0000	11.1250	3.1250	4.2500	0.14	0.06	7.87	10.50	1.41	34400	12200
20	NA94700–94114D	7.0000	11.3750	4.3750	5.6250	0.22	0.06	7.99	10.71	1.25	57000	24000
41	HM237545NA–HM237510D	7.0000	11.3750	4.3750	5.6250	0.22	0.06	7.95	10.68	1.83	67500	19400
42	NA94700–94118D	7.0000	11.7500	4.3750	5.6250	0.22	0.06	7.99	10.71	1.25	57000	24000
43	H239649NA–H239612D	7.3750	12.6250	5.4375	7.3125	0.22	0.06	8.43	11.73	1.83	89000	24300
21	NA93800–93127D	8.0000	12.5000	4.3750	5.7500	0.22	0.06	9.06	11.81	1.12	65800	29400
*22	NA130902–131401D	9.0000	14.0000	4.3750	5.7500	0.25	0.06	10.08	13.01	1.77	72200	20400
44	NA170950–171451D	9.5000	14.5000	3.3750	4.7500	0.25	0.06	10.59	13.27	1.61	51000	15800

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MILITARY STANDARD
MS 19088
SHEET 4 OF 6

FED. SUP CLASS 3110

3110-0519

BEARINGS MANUFACTURED WITH CORRECT RUNNING CLEARANCE FOR THESE FITS.

Dash No.	Live shaft diameter limits	Dead shaft diameter limits		Housing	
	Tight fit	Tight fit	Loose fit	Live shaft bore limits	Dead shaft bore limits
				Loose fit	Tight fit
* 1	0.7515–0.7510	0.7505	0.7500	1.8534–1.8524	1.8494–1.8484
* 2	1.2515–1.2510	1.2505	1.2500	2.3155–2.3145	2.3115–2.3105
* 3	1.5015–1.5010	1.5005	1.5000	2.7200–2.7190	2.7160–2.7150
* 4	1.7515–1.7510	1.7505	1.7500	3.7530–3.7520	3.7490–3.7480
23	1.9700–1.9695	1.9690	1.9685	3.5463–3.5453	3.5423–3.5413
* 5	2.0020–2.0010	2.0005	2.0000	3.6748–3.6738	3.6708–3.6698
6	2.0020–2.0010	2.0005	2.0000	4.2530–4.2520	4.2490–4.2480
24	2.1270–2.1260	2.1255	2.1250	4.3780–4.3770	4.3740–4.3730
25	2.1674–2.1664	2.1659	2.1654	3.9400–3.9390	3.9360–3.9350
* 7	2.3770–2.3760	2.3755	2.3750	4.8780–4.8770	4.8740–4.8730
26	2.7525–2.7515	2.7505	2.7500	4.7274–4.7264	4.7234–4.7224
27	2.7584–2.7574	2.7564	2.7559	4.7274–4.7264	4.7234–4.7224
28	2.8775–2.8765	2.8755	2.8750	5.0030–5.0020	4.9990–4.9980
* 8	3.0025–3.0015	3.0005	3.0000	5.3780–5.3770	5.3740–5.3730
9	3.0025–3.0015	3.0005	3.0000	6.0030–6.0020	5.9990–5.9980
*10	3.5030–3.5020	3.5010	3.5000	6.0030–6.0020	5.9990–5.9980
11	3.5030–3.5020	3.5010	3.5000	6.3780–6.3770	6.3740–6.3730
29	3.7530–3.7520	3.7510	3.7500	6.3780–6.3770	6.3740–6.3730
30	3.7530–3.7520	3.7510	3.7500	7.0896–7.0886	7.0856–7.0846
31	3.7530–3.7520	3.7510	3.7500	7.1280–7.1270	7.1240–7.1230
32	4.0030–4.0020	4.0010	4.0000	6.6280–6.6270	6.6240–6.6230
33	4.0030–4.0020	4.0010	4.0000	7.0896–7.0886	7.0856–7.0846
12	4.0030–4.0020	4.0010	4.0000	7.1280–7.1270	7.1240–7.1230

BEARINGS MANUFACTURED WITH CORRECT RUNNING CLEARANCE FOR THESE FITS.

Dash No.	Live shaft diameter limits	Dead shaft diameter limits		Housing	
	Tight fit	Tight fit	Loose fit	Live shaft bore limits	Dead shaft bore limits
				Loose fit	Tight fit
*13	4.0030–4.0020	4.0010	4.0000	7.5030–2.5020	7.4990–7.4980
34	4.1280–4.1270	4.1260	4.1250	7.0896–7.0886	7.0856–7.0846
35	4.1280–4.1270	4.1260	4.1250	7.1280–7.1270	7.1240–7.1230
14	4.5035–4.5025	4.5010	4.5000	7.5030–7.5020	7.4990–7.4980
15	4.5035–4.5025	4.5010	4.5000	8.3780–8.3770	8.3740–8.3730
36	5.0035–5.0025	5.0010	5.0000	7.1905–7.1895	7.1865–7.1855
*16	5.0035–5.0025	5.0010	5.0000	8.1280–8.1270	8.1240–8.1230
*17	5.0035–5.0025	5.0010	5.0000	9.2530–9.2520	9.2490–9.2480
*18	5.5040–5.5030			9.6280–9.6270	
37	5.6290–5.6280			7.8780–7.8770	
38	5.7540–5.7530			9.5030–9.5020	
19	6.0040–6.0030			10.0030–10.0020	
39	6.8795–6.8785			9.7530–9.7520	
40	7.0045–7.0035			11.1280–11.1270	
20	7.0045–7.0035			11.3780–11.3770	
41	7.0045–7.0035			11.3780–11.3770	
42	7.0045–7.0035			11.7530–11.7520	
43	7.3795–7.3785			12.6310–12.6290	
21	8.0050–8.0040			12.5060–12.5040	
*22	9.0055–9.0045			14.0060–14.0040	
44	9.5060–9.5050			14.5060–14.5040	

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MILITARY STANDARD
MS 19088
SHEET 5 OF 6

FED. SUP CLASS 3110

3110-0519

NOTES:

(A) 1. MATERIAL: Cones, cups, and rollers: bearing quality - carburizing grade alloy or through-hardening grade alloy steel in accordance with FED-STD-66. Cage; carbon steel (stamped). Powdered process steel, e.g., Sinta Forge may be used.

(A) 2. HARDNESS: Cones, cups and rollers: 58 to 64 Rockwell C.

3. TOLERANCES: Standard grade (AFBMA Class 4) (allowable tolerances are in 0.0001 inch).

Cone bore (A) (A)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0	3.0000	5	0
3.0000	12.0000	10	0

Cup diameter (B)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0.0000	12.0000	10	0
12.0000	24.0000	20	0

Bearing width (C) (A)			
Bore size range		Tolerance	
Over	Incl.	Plus	Minus
0	5.0000	100	0
5.0000	12.0000	300	0

Assembled bearing maximum radial runout		
Cup O.D. (B)		Tolerance
Over	Incl.	
0	24.0	20

4. DIMENSIONS: All dimensions are in inches. Dimensions P and S are recommended shaft and housing shoulder diameters. Dimensions R and Y are the maximum fillet radii on the shaft and the housing respectively, which will be cleared by the bearing corners.

5. OPERATING TEMPERATURE: Recommended operating temperature not to exceed 121° Celsius (250° Fahrenheit).

6. LUBRICATION: Bearings shall be furnished without lubrication. Bearings shall be furnished with preservative per MIL-C-11796, Class 3.

(A) 7. BASIC DYNAMIC LOAD RATING: Basic dynamic load rating is that constant stationary load which a group of apparently identical bearings with stationary cups (outer rings) can endure for a rating life of 90 million revolutions of the cone (inner ring). The basic dynamic load ratings listed herein are based on a rated life of 90 million revolutions or 3000 hours at 500 r/min.

To compare the load ratings on this document with others whose basis for rating are other than 90 million revolutions or 3000 hours at 500 r/min use the following formula:

$$C = \text{Other Bearing Rating} \times \left(\frac{R_1}{500}\right)^{1/f} \times \left(\frac{H_1}{3000}\right)^{1/f}$$

where:

R_1 = r/min at which other bearing is rated
H_1 = Hours life at which other bearing is rated
f = Other bearing fatigue factor

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MILITARY STANDARD

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SHEET 6 OF 6

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8. RATING LIFE (HOURS): Rating life is the number of hours at some constant speed of the cone (inner ring) that 90 percent of a group of apparently identical bearings will complete or exceed before first evidence of fatigue develops. The magnitude of the rated life in hours is found from the following:

$$L_{10} = \frac{1.5 \times 10^6}{R} \left(\frac{C}{P}\right)^{10/3} \text{ hours}$$

where:

C = Basic dynamic load rating, lb.
P = Equivalent load (combined radial and thrust load), lb.
R = Revolutions per minute, r/min.

The average life is approximately four times the rating life.

9. K FACTOR: The K factor is the ratio of the basic radial dynamic load rating (for a one-row bearing) to basic thrust dynamic load rating.

10. CONTACT ANGLE: All bearings are normal angle bearings having a contact angle (α) between 10 and 19 degrees. The contact angle is the angle between the line of action of the roller load and a plane perpendicular to the bearing axis.

11. PART NUMBER: The MS part number consists of the MS number, plus the dash number. Example: MS19088-17. The AFBMA (Anti-Friction Bearing Manufacturers Association) cup and cone numbers are for reference only and are not to be used for ordering purposes.

12. CAGE CLEARANCE: Designers should provide a clearance of .125 inch minimum between the outside edge of the cage and the housing counterbore.

13. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

14. Referenced documents shall be of the issue in effect on the date of invitation for bids, or request for proposal.

15. TABLE II, a numerical listing of dash numbers cross-referenced to bore size A, is included as information to assist in locating a known dash number in TABLE I.

TABLE II

Dash No.	Bore A	Dash No.	Bore A	Dash No.	Bore A	Dash No.	Bore A
1	0.7500	12	4.0000	23	1.9685	34	4.1250
2	1.2500	13	4.0000	24	2.1250	35	4.1250
3	1.5000	14	4.5000	25	2.1654	36	5.0000
4	1.7500	15	4.5000	26	2.7500	37	5.6250
5	2.0000	16	5.0000	27	2.7559	38	5.7500
6	2.0000	17	5.0000	28	2.8750	39	6.8750
7	2.3750	18	5.5000	29	3.7500	40	7.0000
8	3.0000	19	6.0000	30	3.7500	41	7.0000
9	3.0000	20	7.0000	31	3.7500	42	7.0000
10	3.5000	21	8.0000	32	4.0000	43	7.3750
11	3.5000	22	9.0000	33	4.0000	44	9.5000

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